CARD 1 OF 20 NET CAPACITY **QUESTION**

You're laying out a **steel suction tank** for a fire pump and a **gravity tank** on a tower. Where does **net capacity** start and stop on each one?

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CARD 2 OF 20 SIZING (ANNEX) **QUESTION**

System needs **1000 gpm for 30 min**. Suction tank is **20 ft in diameter**. The annex adds two allowances on top of the fire volume. What are they, and what **standard size** do you buy?

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CARD 3 OF 20 REFILL TIME **QUESTION**

A **300,000 gal** suction tank — all of it is required fire volume. What's the **minimum fill rate** the water supply must deliver?

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CARD 4 OF 20 TANK TYPES **QUESTION**

Owner wants about **750,000 gal** on site. Which NFPA 22 tank types have a **standard capacity range** that reaches that?

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CARD 2 OF 20 SIZING (ANNEX) **ANSWER**

Fire volume: 1000 × 30 = **30,000 gal**.
20 ft dia → π × 10^2 × 1 ft × 7.48 ≈ **2349 gal per foot** of height.
+ **1 ft** band between fill sensor and overflow → 2349 gal
+ **1.68 ft** left in the bottom (about the suction pipe centerline, so the pump doesn't cavitate) → 3948 gal
30,000 + 6297 = **36,297 gal → 40,000 gal** standard tank.

NFPA 22 (2018) · Ch. 4 (worked example is in Annex A — advisory)

Trap: buying a 30,000 gal tank for a 30,000 gal demand. The mandatory rule is that the tank meets the demand for the full duration; the sensor band and pump-protection water ride on top.

CARD 1 OF 20 NET CAPACITY **ANSWER**

Suction tank: from the **overflow inlet** down to the **level of the vortex plate**.
Every other tank: from the **overflow inlet** down to the **discharge outlet**.
The net volume between those two points has to be **at least the rated capacity**.
Dual-service tank: the fire reserve must be held back automatically — the other-use pipe is separate and draws from **above** the fire-protection level.

NFPA 22 (2018) · Ch. 4 · Ch. 14

Trap: counting water below the vortex plate. On a suction tank the bottom of usable water is the plate, not the tank floor.

CARD 4 OF 20 TANK TYPES **ANSWER**

• **Embankment-supported coated fabric:** 100,000 to 1,000,000 gal, in 100,000 gal steps
• **Prestressed concrete:** 10,000 to 1,000,000 gal
Topping out lower: welded steel list ends at **500,000** · bolted steel ~4000 to ~**500,000** · reinforced concrete 10,000 to **500,000** · wood to **100,000**.
Allowed materials: steel, wood, concrete, coated fabric, fiberglass (FRP).

NFPA 22 (2018) · Ch. 4, 5, 6, 8, 9, 10

Trap: treating the standard list as a cap. Other sizes are permitted — a 750,000 gal welded steel tank is legal, it just isn't a listed standard size.

CARD 3 OF 20 REFILL TIME **ANSWER**

The supply has to refill the minimum required fire-protection volume in **8 hours or less**.
8 hr × 60 = 480 min → 300,000 ÷ 480 = **625 gpm**.
The fill pipe, a filling pump, and a bypass fill are all sized to that same 8-hour rule.

NFPA 22 (2018) · Ch. 4 · Ch. 14

Trap: pressure tanks are different. Their fill supply must put the water back *and* restore the air pressure in **4 hours** or less (Ch. 7).

sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 20 PRESSURE TANK (ANNEX) **QUESTION**

Light hazard: **125 gpm for 30 min**, **55 psi** needed at the tank discharge flange. You pick a **7000 gal** pressure tank. What air pressure does it hold?

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CARD 6 OF 20 PRESSURE TANK PIPING **QUESTION**

Draw the three pipes on a **pressure tank**: discharge, fill, air. Minimum size of each, and which valves go on them?

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CARD 7 OF 20 PRESSURE TANK ROOM **QUESTION**

The pressure tank goes in its own room. What **clearances, heat and housing** do you show — and what limits where a pressure tank can serve at all?

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CARD 8 OF 20 DISCHARGE PIPE **QUESTION**

Elevated gravity tank on its own tower feeds the yard main. Minimum **discharge pipe** size, and which valves go where?

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CARD 6 OF 20 PRESSURE TANK PIPING **ANSWER**

Discharge: sized for the system, never under **4 in.**; fitting sticks up **2 in.** off the bottom as a settling basin; listed **horizontal swing check + listed indicating valve** at the tank.
Fill: at least **1½ in.**, run on its own; listed check + indicating control valve, horizontal, near the tank, with the **indicating valve between the check and the tank**.
Air: at least **1 in.**, entering **above the water level**; swing check + globe valve, globe valve on the tank side.

NFPA 22 (2018) · Ch. 7

Trap: gravity and suction tank discharge minimum is **6 in.** (Ch. 14). Pressure tank is **4 in.** Don't swap them.

CARD 5 OF 20 PRESSURE TANK (ANNEX) **ANSWER**

Water: 125 × 30 = **3750 gal** → 3750 ÷ 7000 ≈ 54% water, **46% air** (A = 0.46).
$P_i = (P_f + 15) \div A - 15$
$= (55 + 15) \div 0.46 - 15 \approx$ **137 psi**.
The mandatory rule behind it: enough air and pressure to push **all** the water out and still hold the needed residual at the top of the system.

NFPA 22 (2018) · Ch. 7 (formula + example are in Annex A — advisory)

Trap: same system in a **5000 gal** tank at 75% water: 70 ÷ 0.25 − 15 = **265 psi** — past the 175 psi most parts are rated for. Too small a tank drives pressure up. Tell the owner the fill point; overfilling steals air.

CARD 8 OF 20 DISCHARGE PIPE **ANSWER**

- Sized off the hydraulic demand, but never under **6 in.**
- Listed **check valve, horizontal**, in a **pit under the tank**
- Listed **indicating control valve on each side** of the check; the downstream one sits between the check and any branch to other piping
- Independent tower: that valve goes in the pit with the check, preferably yard side of the base elbow
- Settling basin in the tank bottom: **4 in.** flat bottom · **18 in.** suspended bottom

NFPA 22 (2018) · Ch. 14

Trap: a missing check isn't always wrong. A **suction tank** whose discharge isn't also its fill doesn't need one. A gravity tank does.

CARD 7 OF 20 PRESSURE TANK ROOM **ANSWER**

- Free access to every connection and manhole, and at least **18 in.** around the rest of the tank
- At least **3 ft** from the floor to any part of the tank
- Room heated to **40°F** min; watertight floor that drains outside; noncombustible housing where it could freeze
- Add sprinklers when anything combustible in or around the tank room could heat the tank
- Service: **limited** systems (NFPA 13, 14, 15 types); **capacity is the AHJ's call**

NFPA 22 (2018) · Ch. 7

Don't forget the compressor: at least **16 cfm** free air for a 7500 gal tank, **20 cfm** for larger. It goes in the tank house unless the AHJ allows elsewhere.

sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 9 OF 20 ANTI-VORTEX PLATE **QUESTION**

Suction tank with a **10 in. suction outlet**. Size the **anti-vortex plate** and set its height off the tank floor.

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CARD 10 OF 20 FILLING **QUESTION**

Fire pump suction tank, filled automatically off the city. Where does the **fill pipe discharge** inside the tank — and what if it comes **over the top**?

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CARD 11 OF 20 OVERFLOW **QUESTION**

Fill line is **4 in.** and nobody knows the max fill rate. Size the **overflow**.

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CARD 12 OF 20 BREAK TANK **QUESTION**

Fire pump rated **1000 gpm** takes suction from a **break tank**. Minimum tank size? And what makes it a break tank instead of a suction tank?

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CARD 10 OF 20 FILLING **ANSWER**

• Automatic fill discharges into the **opposite half** of the tank from the pump suction
• Over-the-top fill: outlet **turned down**
• A permanent fill pipe is required (an approved manual-refill plan only where no permanent supply exists)
• Keep it full: never more than **4 in.** below the fire-service level
• Filling from the system through a **bypass around the check**: at least **2 in.**, with a listed indicating valve kept **closed** except while filling

NFPA 22 (2018) · Ch. 14

Trap: fill and suction on the same side. Keep the incoming flow away from the pump suction — opposite half, every time.

CARD 9 OF 20 ANTI-VORTEX PLATE **ANSWER**

• Required on the discharge outlet of **every** suction tank
• Horizontal steel plate at least **2× the outlet diameter** → **20 in.** min
• Height off the floor = **½ the discharge pipe diameter** → 5 in. … but **6 in.** is the minimum → **set it at 6 in.**
• Suction tank control valve: **OS&Y**, located per NFPA 20
Annex: a standard **48 × 48 in.** plate is preferred — it covers every pump suction size normally used.

NFPA 22 (2018) · Ch. 14

Trap: the ½-diameter rule loses to the 6 in. floor on any pipe under 12 in. And the plate is the bottom of your **net capacity** (card 1).

CARD 12 OF 20 BREAK TANK **ANSWER**

Break tank = feeds a fire pump but holds **less than the full fire demand** (flow × duration).
Minimum size: **15 min** with the pump at **150%** of rated.
1000 × 1.5 = 1500 gpm × 15 min = **22,500 gal** min.
Annex reasons to use one: a break from the city (backflow), steady pump suction pressure, topping up a weak city supply, no room for a full-demand tank.

NFPA 22 (2018) · Ch. 3 · Ch. 4 · Ch. 14

Trap: sizing off the sprinkler demand. The minimum comes from the **pump** (150% of rated), not the system flow.

CARD 11 OF 20 OVERFLOW **ANSWER**

• Capacity must be greater than the fill connection, and never under **3 in.** anywhere along it
• Fill rate unknown → go **one pipe size up from the fill** or more, with an inlet (concentric reducer or equal) at least **2 in. larger** in diameter
• Inlet at the **top capacity line**, nothing blocking the flow
• Embankment (coated fabric) tank: sized for the fill rate, or **2 in. larger** than the fill line

NFPA 22 (2018) · Ch. 14 · Ch. 9

Trap: the overflow inlet is the **top of net capacity** — move it and you change the tank's rated volume. And the overflow does **not** count as roof vent area (Ch. 4).

sheet 3 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 13 OF 20 BREAK TANK REFILL **QUESTION**

Break tank holds **30,000 gal**. System demand is **1000 gpm for 60 min**. Which refill rules apply, and at what rate?

CARD 14 OF 20 TANK HEATING **QUESTION**

Tank in a cold climate. What **water temperature** does the heater hold, where's the alarm set, and what weather do you design to?

CARD 15 OF 20 FROST PROTECTION **QUESTION**

Site's lowest one-day mean temperature is **15°F**. The tank piping runs exposed. What do you have to show?

CARD 16 OF 20 SUPERVISION **QUESTION**

The spec says the tank is **supervised**. List the signals NFPA 22 calls for — with the numbers.

CARD 14 OF 20 TANK HEATING **ANSWER**

- Any tank subject to freezing **must be heated**
- Coldest water in the tank or riser held at **42°F or above** in the coldest weather
- **Low water temperature alarm at 40°F**
- Design weather = **lowest mean temperature for one day**, from an approved source
- Heater sized from the Ch. 16 heat-loss chart; heat loss calcs go in the submittal

NFPA 22 (2018) · Ch. 16 · Ch. 4

42 holds, 40 alarms. Valve pits, valve houses and heater houses only need 40°F (Ch. 15). Swapping which number goes where is the easy miss.

CARD 13 OF 20 BREAK TANK REFILL **ANSWER**

Tank = 30 min of max demand (1000 × 30) → the **30-min-or-more** path:
Refill = 110% × (Total demand − Tank) ÷ Duration
= 1.10 × (60,000 − 30,000) ÷ 60 = **550 gpm**. Manual bypass: same rate.
Fill valve kept at **40°F** min; auto fill starts no more than **4 in.** below the overflow.
Under 30 min: two auto refill lines, *each* at **150%** of pump capacity (or 110% of max system flow if supply can't), a 150% manual bypass, and a local low-level alarm.

NFPA 22 (2018) · Ch. 14

Trap: one fill line on an under-30-min tank. It takes **two**. Either path, the refill must be **listed and automatic**.

CARD 16 OF 20 SUPERVISION **ANSWER**

- Water temp **below 40°F**, and its return to 40°F
- Water level down **12 in.** on gravity/suction tanks — **3 in.** on a pressure tank — and its return to normal
- Pressure tank only: **10 psi below** and **10 psi above** normal

Provide for sensors per NFPA 72: two temperatures, two levels, two pressures (pressure tanks only).

NFPA 22 (2018) · Ch. 14

Trap: that 12 in. level band is the same band the annex adds to tank size (card 2). Leave it out and a low-level signal is already eating your fire volume.

CARD 15 OF 20 FROST PROTECTION **ANSWER**

- Lowest one-day mean **20°F or lower** → a **listed frostproof casing** around all exposed tank piping. 15°F qualifies.
- Also protect piping in unheated buildings that drop below freezing, and dry risers of elevated tanks
- Valve pit, valve house, heater house: heated to at least **40°F** in the worst weather
- Large steel-plate risers **3 ft**+ in diameter can go without a casing **if properly heated**

NFPA 22 (2018) · Ch. 15 · Ch. 14

Trap: using the record low. The trigger is the lowest **mean temperature for one day** — the same number that drives tank heating.

CARD 17 OF 20 FRP — BURIED **QUESTION**

Customer wants a **fiberglass (FRP) tank buried under a parking lot**. What cover and connections do you show?

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CARD 18 OF 20 FRP — INDOORS **QUESTION**

A **horizontal FRP tank, 6 ft in diameter**, sits on cradles **2 ft** off the pump room floor. What does that do to your sprinkler layout?

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CARD 19 OF 20 LOCATION / EXPOSURE **QUESTION**

Elevated tank's steel tower sits **15 ft** from a combustible building with windows facing it. What does the tower steel need?

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CARD 20 OF 20 FOUNDATIONS **QUESTION**

Plot plan: a **steel suction tank on a ringwall** and an **elevated tank on piers**. What elevations and dimensions do you check?

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CARD 18 OF 20 FRP — INDOORS **ANSWER**

Two things:

• An FRP tank inside a building is protected by sprinklers designed as **Ordinary Hazard Group 2** — or per NFPA 13 if the room's hazard is higher than OH2

• A horizontal FRP tank **over 4 ft** in diameter and **18 in. or more** off the floor falls under the NFPA 13 **obstruction** rules. 6 ft dia at 2 ft up → **both** triggers met.

NFPA 22 (2018) · Ch. 11

Trap: treating the tank as just a thing to sprinkle over. At that size it's an **obstruction** — you may need heads under it.

CARD 17 OF 20 FRP — BURIED **ANSWER**

• FRP is for water at **atmospheric pressure only**

• **Traffic:** 36 in. of backfill — or 18 in. compacted backfill **plus** 6 in. reinforced concrete or 9 in. asphalt; paving runs **12 in.** past the tank outline

• **No traffic:** at least 12 in. compacted backfill (or 12 in. + a 4 in. reinforced slab)

• Under railroad tracks: **4 ft** of cover · whole tank below the frost line

• Vent above ground, **2 in.** min · level readable above ground · fill accessible above ground

• Fire department cistern: lowest fire water no more than **15 ft** below where the engine sets up

NFPA 22 (2018) · Ch. 11

Trap: cover is measured from the **top of the tank to finished grade** — account for future grade.

CARD 20 OF 20 FOUNDATIONS **ANSWER**

Suction tank: concrete slab, or concrete ringwall with a sand / crushed stone / granular cushion

• Ringwall at least **10 in.** wide, reinforced, right under the shell, below the frost line **and** at least **1 ft** below finished grade

• Ring top **6 in. min to 12 in. max** above surrounding grade

• Sand cushion on a ringwall: at least **3 in.**

Elevated tank piers: tops at least **6 in.** above grade; bottoms below frost line or **4 ft** down, whichever is deeper

Concrete: **3000 psi** min

NFPA 22 (2018) · Ch. 12

Trap: the ring top has a **maximum** too. A pad set 18 in. proud of grade is out.

CARD 19 OF 20 LOCATION / EXPOSURE **ANSWER**

• Steel supporting a tank **within 20 ft** of exposures, combustible buildings, or windows/doors fire could come out of → **fireproofed**

• Fireproofing rating: **2 hours** min

• No yard room to get clear → fireproof the steel **or** protect it with open sprinklers

• Supports inside a building near combustibles: fireproofed **6 ft** above combustible roof coverings, and within 20 ft of windows/doors

• **Interior timber** may never support or brace a tank structure

NFPA 22 (2018) · Ch. 4

Trap: forgetting the building. A tank or trestle on a roof means the **building** has to be designed for the full load.

sheet 5 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back